

Where the Data Came From

Unit 5: Causal Claims, Selection, and What You Are Allowed to Conclude

Stat 220 | Statistics for Data Science

Where this fits

- Units 3 through 4 built models, judged them honestly, and put error bars on what they predict. All of that assumed the rows in front of you were the rows you should have.
- This unit asks the two questions that come before any of it. Where did these rows come from, and does the pattern in them mean what it looks like?
- Those turn out to be one question. A confounder and a biased sample are both cases of the data being assembled in a way that plants a pattern you did not put there.

Principle: the order that matters

No amount of modeling fixes a dataset that answers a different question than the one you asked. The check happens first, or it does not happen.

What this unit gives you

Things to know

- what a causal effect is, in terms of what would have happened otherwise
- why randomization works, and exactly what it buys
- the three roles a third variable can play, and what to do about each
- that colliders and survivorship bias are one mechanism
- population, sampling frame, and the designs worth knowing
- how data goes missing, and what that does to your answer

Mistakes to stop making

- controlling for everything you have
- reading a slope from observational data as an effect
- treating sample size as evidence of representativeness
- studying only the survivors, winners, or responders
- dropping missing rows without asking why they are missing
- accepting a proxy as the thing you meant to measure

An effect is a comparison with something that never happened

- For one customer, the effect of the discount is: what they spend *with* it, minus what they would have spent *without* it.
- You can only ever observe one of those two. The other is a counterfactual, and it is missing by definition.
- So causal inference is a **missing data problem**: everything hinges on how you fill in the world that did not happen.

Principle: the question behind every causal claim

“Compared to what?” If someone cannot tell you what their comparison group is, they do not have a causal estimate.

Why randomization is the gold standard

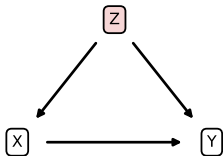
- Randomly assigning treatment makes the two groups comparable *in expectation* on everything: age, motivation, prior spending, and every variable you never thought to measure.
- That last clause is the important one. Adjustment can only handle confounders you measured. Randomization handles the ones you did not.
- What it does *not* fix: small samples, dropout after assignment, non-compliance, contaminated control groups, or measuring the wrong outcome.

In practice: say this precisely

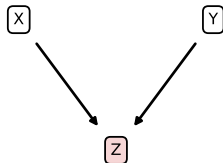
“Randomization does not make the groups identical. It makes any difference between them the kind of difference we can quantify with a standard error.”

The three roles a third variable can play

CONFOUNDER: control for it



COLLIDER: never control for it



MEDIATOR: control only if you want the direct effect



The arrows are what you believe about the world. Which variables to control for is decided by these diagrams, *not* by which ones improve the fit.

Confounders: the classic problem

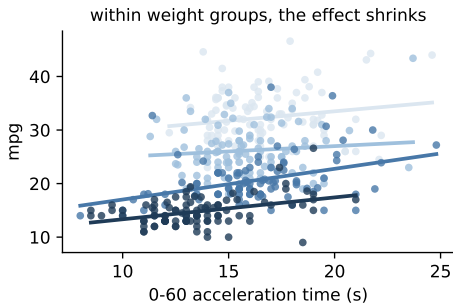
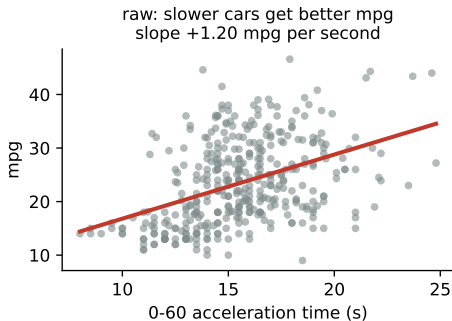
- A **confounder** causes both the treatment and the outcome, producing an association with no causal path behind it.
- Ice cream sales and drownings both rise with temperature. Neither causes the other.
- In business: customers who *choose* the premium plan were already more engaged, so premium looks like it causes retention.

Principle: control for confounders, and say which ones you could not measure

The honest sentence is: “adjusting for tenure, plan size, and industry, the difference is X, but I cannot rule out that more motivated teams chose the feature.”

Confounding in real data

Weight drives both acceleration and fuel economy



392 cars. Slower-accelerating cars average **1.20 mpg better per extra second** to 60. Once you adjust for weight, which drives both, the association collapses to **0.26**. Nearly 80% of the raw relationship was weight, relabeled.

Colliders

- A **collider** is caused by both variables. Conditioning on it creates an association where none existed, exactly as the dice demonstrated.
- Selecting your sample on a collider is the same mistake: analyzing only customers who *contacted support*, only patients who were *admitted*, only startups that *got funded*.
- This is why “add every variable you have to the regression” is bad advice. Some of those variables are colliders, and adding them injects bias rather than removing it.

Trap: the kitchen-sink regression

More controls is not safer. Each control is a claim about the causal diagram, and the wrong one makes your estimate worse in a direction you cannot sign.

Mediators

- A **mediator** sits on the causal path: training raises skill, skill raises sales.
- If you regress sales on training *controlling for skill*, you have removed the very channel through which training works, and the effect vanishes.
- Controlling for a mediator answers a different, narrower question: is there a direct effect *other than* through skill?

In practice: a question that trips people up

“Should you control for everything?” The right answer is no, and the reason is exactly this: confounders yes, colliders never, mediators only if you deliberately want the direct effect.

The rules, in one table

Role	Structure	What to do
Confounder	causes both X and Y	Control for it. Omitting it biases the estimate.
Collider	is caused by both X and Y	Never control for it , and never select the sample on it.
Mediator	X causes it, it causes Y	Leave it out for the total effect.

Principle: the order of operations

Decide the role from what you believe about the world, *then* write the model. You cannot read the role off a correlation matrix, and the fit will not tell you either.

Your turn #1

Your turn: classify the variable

You want the effect of a **new onboarding flow** on 90-day retention. For each variable, say whether you should control for it, and why:

- 1 Company size at signup.
- 2 Number of features used in week one.
- 3 Whether the customer contacted support at any point.

Your turn #1: answer

- **Company size:** a confounder. It affects both which flow they got (if assignment was not random) and retention. **Control for it.**
- **Features used in week one:** a mediator. That is how onboarding works. **Do not control** unless you specifically want the effect that runs through some other channel.
- **Contacted support:** a collider. It is caused by both the onboarding experience and by the customer's underlying trouble. **Never control for it**, and do not restrict the sample to those who contacted support.

Principle: draw the arrows before you write the formula

Two minutes with a diagram prevents a month of arguing about a coefficient.

Colliders and survivorship bias are the same thing

You have now met the collider: conditioning on a common consequence of two variables makes them look related when they are not. That mechanism has other names depending on who got filtered out.

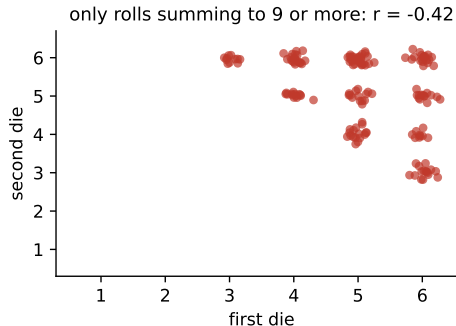
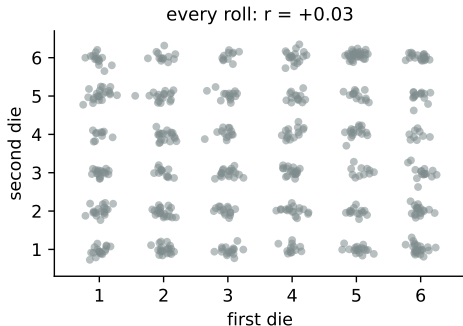
Name	What got filtered	The pattern it plants
Collider bias	cases that scored high on a shared consequence	two unrelated causes look negatively related
Survivorship bias	anything that did not last	the traits of survivors look like the causes of survival
Nonresponse bias	people who chose not to answer	the answers you got look like the answers everyone would give

Principle: one question covers all three

Ask who is missing from this table, and what it took to get in. That single question catches every row above.

Selection can create a relationship

Two independent things, made dependent by selection



Independent to begin with. Among high-sum rolls, a low first die *requires* a high second die, so the two become strongly negatively related. Nothing about the dice changed. We changed who was in the dataset.

Where this shows up in real data

- Restaurants you hear about are either great food or great location, because bad on both close. So in your data, food quality and location look negatively related.
- Admitted students are either strong on tests or strong on essays. Within the admitted class, the two look like they trade off.
- Hospitalized patients have some serious condition, so among them, unrelated diseases appear connected.

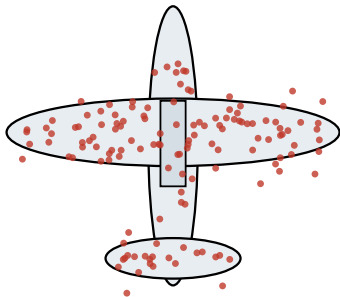
Principle: the sample is part of the analysis

Before you interpret any relationship, ask who got into the dataset, and what it took to get in.

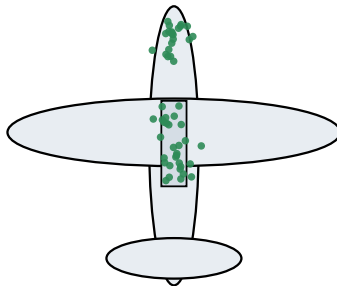
Survivorship bias

The holes you never see are the ones that matter

Damage on planes that came back

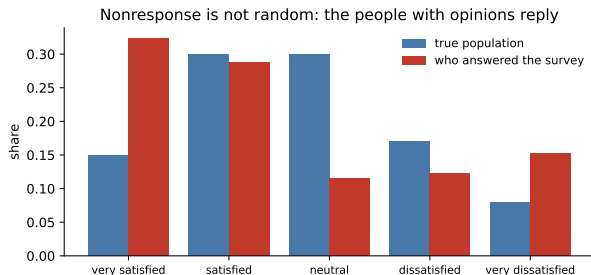


Where armor was actually added



In World War II, analysts studied bullet holes in returning bombers and proposed armoring the wings, where the holes were. Abraham Wald pointed out the obvious once you see it: those planes came back. Armor belongs where the returning planes have *no* holes, because planes hit there did not return.

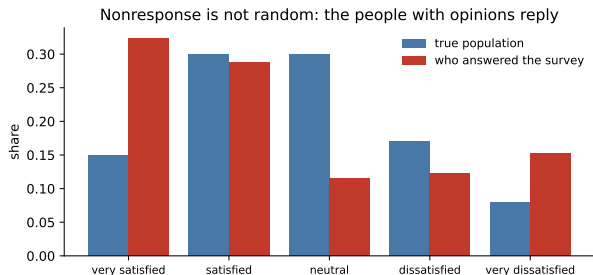
Nonresponse



People with strong feelings answer surveys. The indifferent middle does not.

- Your satisfaction data becomes bimodal even when the population is not.
- Response rates for phone polls have fallen from over 30% to under 10%, and the tilt is what makes modern polling hard.

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Weighting can correct for *measured* differences. It cannot correct for people who differ in ways you did not record.

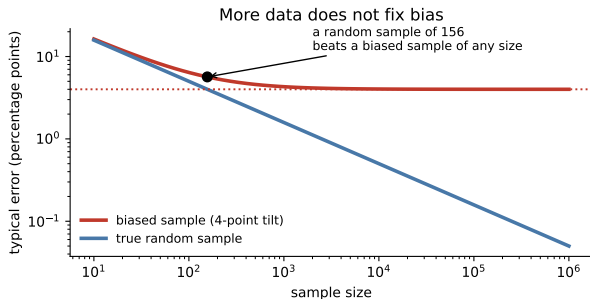
Two famous failures

- **1936:** the *Literary Digest* mailed 10 million ballots and got 2.4 million back, predicting a Landon landslide. Roosevelt won 46 states. The frame came from car and telephone registries in the Depression, and only the enthusiastic mailed back.
- At the same time George Gallup polled about 50,000 people, chosen to match the population, and got it right.
- **The lesson repeats:** 2.4 million biased responses lost to 50,000 designed ones.

Principle: design beats volume

Sampling error shrinks with n . Sampling bias does not shrink at all.

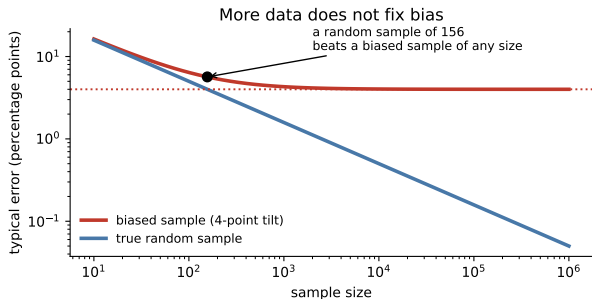
The big-data paradox



A sample with a 4-point systematic tilt never gets more accurate than 4 points, however many rows you add.

- A *random* sample of about **156** matches it.
- Worse: the huge sample reports a tiny standard error, so it is confidently wrong.

The big-data paradox



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- A *random* sample of about **156** matches it.
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This is exactly how 40 million app users can mislead you about your market.

Population, frame, sample

- **Population:** everyone you want to describe. All customers, all flights, all voters.
- **Sampling frame:** the list you can actually draw from. Your CRM, the phone book, people with app notifications enabled.
- **Sample:** who ended up in the data.

Bias enters at both arrows: frame does not match population (**undercoverage**), and sample does not match frame (**nonresponse**).

Principle: name your frame out loud

“Our customers” usually means “customers with an email on file who opened the message.” Saying the real frame is often enough to spot the problem.

Four sampling designs

- **Simple random sample:** everyone equally likely. The benchmark.
- **Stratified:** split into groups (region, plan tier), sample within each. More precision for the same cost, and guarantees small groups appear.
- **Cluster:** sample whole groups (stores, classrooms) and measure everyone inside. Cheaper, but the effective sample size is closer to the number of clusters.
- **Systematic:** take every k th record. Fine unless the list has a hidden periodicity.

Trap: cluster samples analyzed as if they were random

Surveying 20 stores with 50 customers each is not 1,000 independent observations. Treating it as such is the standard-error error from Unit 8, at scale.

Your turn #2

Your turn: find the frame problem

A streaming service estimates “average weekly viewing hours per subscriber” from telemetry sent by its smart-TV app.

With a neighbor: name three groups of subscribers this misses or distorts, and say which direction the estimate is biased.

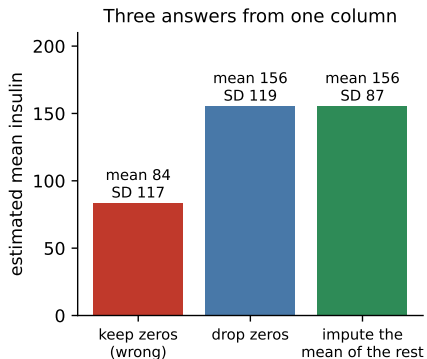
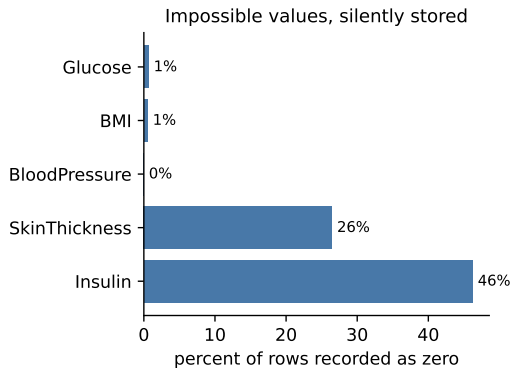
Your turn #2: answer

- Subscribers who watch on phones, laptops, or game consoles are missing entirely, and their habits differ.
- Households sharing one account look like one heavy viewer, while a subscriber who never opens the app sends no telemetry at all and may vanish from the denominator.
- Anyone with telemetry disabled, or an older TV, is excluded, and both correlate with age and income.

Principle: ask what the instrument can see

Telemetry measures device activity, not people watching. The gap between those two is where the bias lives.

Missing values in real data



733 patients. Insulin is recorded as zero for **46%** of them, which is biologically impossible: those are missing values recorded as zero. Keeping them halves the estimated mean. Mean imputation gets the mean right and shrinks the SD from 119 to 87, manufacturing precision that does not exist.

Three ways data goes missing

- **Missing completely at random:** the machine failed randomly. Dropping rows costs precision but is unbiased. Rare in practice.
- **Missing at random (given what you measured):** older patients skip a test more often, and you recorded age. Dropping is biased. Modeling with age is not.
- **Missing not at random:** the value itself drives the absence. High earners refuse to report income, and the sickest patients miss follow-up. Nothing in the data fixes this. You need external information or an explicit assumption.

Trap: silent listwise deletion

Most software drops incomplete rows without telling you. If the dropped rows differ systematically, your entire analysis quietly changed populations.

What to actually do

- 1 **Count and map the missingness** first, per column, and check whether missing rows differ from complete ones on the outcome.
- 2 **Look for disguised codes:** 0, -1 , 999, “N/A”, or an empty string.
- 3 **Add a missingness indicator** when absence itself might be informative. Often it is the most predictive feature you have.
- 4 **Prefer multiple imputation** to single mean imputation, so the extra uncertainty shows up in the interval instead of vanishing.
- 5 **Say what you did**, and show the result both ways.

Weighting

When your sample's composition is wrong but you *know* the true composition, you can reweight.

Age group	% of population	% of respondents	weight
18–34	30%	12%	2.50
35–54	35%	33%	1.06
55+	35%	55%	0.64

- Each young respondent now counts for 2.5 people, so the estimate reflects the population mix rather than who answered.
- It only fixes imbalance on variables you *measured*, and heavy weights inflate the variance.

Trap: weighting as a blank check

“We weighted the survey” does not mean the bias is gone. It means the bias on the weighting variables is gone.

You measure a proxy, not the concept

- “Engagement” becomes clicks. “Teacher quality” becomes test scores. “Customer happiness” becomes a survey score from the 6% who responded.
- Every proxy has a gap between it and the real concept, and decisions get made in that gap.
- **Goodhart’s law:** once a proxy becomes a target, people optimize the proxy, and the gap widens on purpose.

In practice: a strong thing to raise unprompted

“Before I model this, what exactly does this column measure, who recorded it, and when? Half the projects I have seen went wrong at that step.”

Measurement problems that look like findings

- **Instrument drift:** a logging change in March makes April look like a behavior shift.
- **Question wording:** “Do you support banning X?” and “Do you oppose allowing X?” give different numbers from identical people.
- **Social desirability:** people over-report voting, exercise, and reading. They under-report drinking and screen time.
- **Definition changes:** a churn spike caused by moving the churn cutoff from 90 to 60 days.

Trap: check the instrument before the world

Before explaining a sudden jump with a story about the world, check whether the instrument changed on that date. It usually did.

Your turn #3

Your turn: whose data is it?

A hiring team builds a model to predict "good employees" using performance ratings from the past ten years of hires.

With a neighbor: identify one selection problem, one measurement problem, and one missing-data problem in that setup.

Your turn #3: answer

- **Selection:** you only observe performance for people who were *hired*. Rejected candidates who would have excelled are invisible, so the model learns the old process's preferences.
- **Measurement:** “performance rating” is a manager's judgment, carrying whatever biases and grade inflation that manager had. It is a proxy, not the concept.
- **Missing:** people who left early have no rating, and they left for reasons related to the outcome. That is missing not at random.

Principle: the model inherits the process that made the data

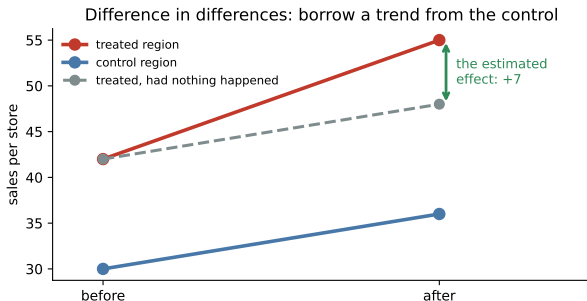
A model trained on past decisions predicts the past decisions, including their mistakes. That is not a bug you can regularize away.

When you cannot randomize

Often you cannot randomize: it is unethical, illegal, too slow, or the change already happened. Four workhorse designs, each buying credibility with an assumption:

- **Matching / adjustment:** compare treated units to similar untreated ones.
Assumes you measured every confounder.
- **Difference in differences:** compare changes over time against a control group.
Assumes the two groups would have moved in parallel.
- **Regression discontinuity:** compare units just above and just below a cutoff.
Assumes units cannot precisely manipulate their position.
- **Instrumental variables:** use a nudge that affects treatment but nothing else.
Assumes the instrument has no other path to the outcome.

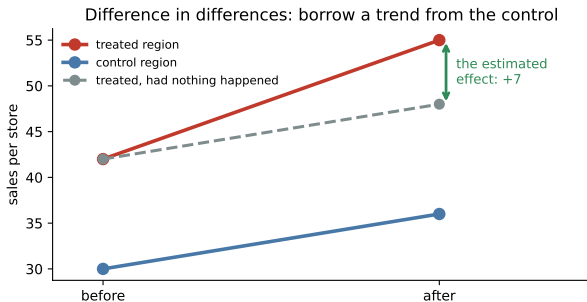
Difference in differences, drawn



The treated region rose by 13. But the control region rose by 6 without any treatment, so the economy was moving anyway.

- Estimated effect: $13 - 6 = 7$.
- Everything rests on the dashed line, the claim that the treated region would have followed the control's trend.

Difference in differences, drawn



So the first thing to check is whether the two lines were parallel *before* the change.

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Difference in differences, with the actual arithmetic

A chain raises pay in its **Ohio** stores in July and leaves **Indiana** alone.

	before	after	change
Ohio (treated)	42.0	55.0	+13.0
Indiana (control)	30.0	36.0	+6.0
difference in differences			+7.0

- Ohio rose 13, but 6 of that would have happened anyway. The estimated effect is **7**.
- It all rests on the claim that Ohio *would have* risen by 6 too, which is untestable after treatment.

Principle: how to check what you cannot test

Plot both series for several periods *before* the change. If they moved together then, the assumption is credible.

Reverse causation, and how to rule it out

- Hospitals with more staff have higher mortality. Do nurses kill people, or do sicker patients get more nurses?
- Ad spend and sales move together, partly because budgets are *set* as a percentage of expected sales.
- Police numbers and crime, tutoring and grades, maintenance and failures: in each case the outcome plausibly causes the treatment.

Principle: the fix is timing, not statistics

Establish that the cause moved *first*, and for a reason unrelated to the outcome. That is what a rollout date, a policy change, or a budget freeze gives you, and no regression can substitute for it.

Natural experiments

- **John Snow, 1854:** two water companies served interleaved houses on the same London streets, one drawing from sewage-contaminated water. The houses were otherwise alike, so the cholera difference was as good as randomized.
- **Minimum wage, 1994:** Card and Krueger compared fast-food employment in New Jersey after a wage increase against neighboring Pennsylvania, a difference in differences that reversed the textbook prediction.
- **Scholarship cutoffs:** students just above and just below a score threshold are nearly identical, so the cutoff acts like a coin flip.

Principle: look for the accident

The best observational studies find a moment where something outside the actors' control assigned the treatment for them.

Your turn #4

Your turn: design the study

Your company rolled out a new sales-training program. Reps who completed it sold 18% more last quarter than reps who did not. Leadership wants to make it mandatory. With a neighbor: name two reasons that 18% may not be the effect of training, and design a study you could actually run next quarter.

Your turn #4: answer

- **Selection:** reps who volunteered were more motivated, and would have improved anyway. **Regression to the mean:** if struggling reps were assigned, expect a bounce with no cause.
- **Timing:** the quarter may have been better for everyone, which a simple before-after comparison would credit to training.
- **The study:** randomize the next cohort. If that is impossible, use a staged rollout by region and compare with difference in differences, checking that the regions were parallel beforehand.

Principle: staged rollouts are free experiments

If you must launch to everyone eventually, launch in a random order. It costs nothing and converts a guess into an estimate.

LLM check: mistake or not? (1 of 2)

LLM check: we asked an LLM about a feature launch

We told it: “Customers who use our new dashboard renew at 91%, others at 68%. What is the effect of the dashboard on renewal?”

It answered: “The dashboard increases renewal by 23 percentage points. To be thorough I would control for every variable available in your database to remove any confounding.”

Mistake, or no mistake?

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Mistake, or no mistake?

Trap: verdict: mistake, twice

First, the 23 points is a comparison between customers who *chose* the dashboard and those who did not, so it is mostly selection. Second, “control for everything” is wrong: some of those variables are colliders and mediators, and controlling for them adds bias.

LLM check: mistake or not? (2 of 2)

LLM check: we asked an LLM about sample size

We told it: “Our survey of app users got 40,000 responses out of 2 million invitations. Is that enough to represent our user base?”

It answered: “Yes, 40,000 is an excellent sample size. The margin of error is under 0.5%, so you can be very confident these results represent your users.”

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Mistake, or no mistake?

Trap: verdict: mistake

A 2% response rate is the headline, not the 40,000. The margin of error describes sampling error only, and says nothing about the 98% who did not answer. The tiny stated error makes the result more dangerous, not more trustworthy.

How to answer “does X cause Y?”

- 1 **State the intervention precisely.** Who gets what, instead of what?
- 2 **Name the comparison group.** If there is none, stop.
- 3 **Draw the diagram:** candidate confounders, colliders, mediators.
- 4 **Ask how treatment was assigned.** Randomized, chosen, or assigned by a rule?
- 5 **Choose the design** that fits the assignment mechanism.
- 6 **Check the design's assumption** explicitly (parallel pre-trends, balance, no manipulation at the cutoff).
- 7 **Report the effect with its uncertainty and its caveats**, including the confounder you could not measure.

An audit to run first

- 1 **Name the population** the decision concerns, and the frame the rows actually came from.
- 2 **Ask who is missing because of the outcome** you are studying.
- 3 **Count the missing values per column**, and look for them in disguise: 0, -1, 999, empty strings.
- 4 **Compare rows with missing values to complete rows** on the outcome. If they differ, dropping them changes the population.
- 5 **Ask what each column really measures**, who recorded it, and when.
- 6 **Check for breaks over time** in the instrument, the definition, or the pipeline.
- 7 **Write down what you could not fix**, and put it in the report.

The traps, all in one place

Trap: the ones that bite most often

- Reading a slope from observational data as an effect.
- Controlling for everything available, which is how you condition on a collider.
- Controlling for a mediator and then reporting the effect as if it vanished.
- Treating a large sample as a representative one.
- Studying only survivors, winners, or people who responded.
- Dropping rows with missing values without asking why they are missing.
- Imputing the mean and then quoting the narrower interval that results.
- Accepting a proxy as the concept you actually meant.

What you should be able to do with software

Nobody is asking you to memorize syntax. You are expected to know *which* procedure the question calls for, what to hand it, and what the output means. With an AI assistant open, you should be able to produce any of these and defend the result.

You should be able to	What you would reach for
Adjusted comparison	<code>smf.ols(ŷ ~ treat + covariates; data)</code>
Estimates within strata	<code>pd.qcut</code> then <code>groupby</code>
Difference in differences	four group means, then the difference of the differences
Check parallel pre-trends	plot both series for several periods before treatment
Randomized assignment for a rollout	<code>rng.integers(0, 2, n)</code> on the unit of assignment
Simulate a confounder or a collider	generate the data yourself and compare estimates
Logistic version for a binary outcome	<code>smf.logit</code>

If you cannot say what the output means, the fact that you produced it is worth nothing.

Where we go next

- That closes Part I. You can build a model, judge it honestly, say how wrong a prediction might be, and now ask whether the data ever supported the claim.
- Part II goes back and earns the machinery. **Unit 6** starts with probability, which is what the standard errors, the p -values, and the distributions have been resting on the whole time.

Principle: the through-line

Part I was about what you can conclude. Part II is about why those tools are entitled to conclude anything at all.